Research ethics and consent

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Research ethics and consent before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dataset and benchmark limitations

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Effect size and practical importance

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Research ethics and consent trace and record the first step you can explain without help.

Use W2: Trace the worked case for Research ethics and **consent**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Research ethics and consent explanation against the supplied facts before discussing it.

Use the worked case for Research ethics and **consent**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Research ethics and consent facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 Trace the worked case

> Trace the worked case. Apply the supplied Research ethics and consent facts and keep your reasoning.

Proposal automatically copies student essays into a study because they were submitted for grading. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Research ethics and consent facts and keep your reasoning.

A report omits this qualification: This classroom policy exercise does not determine real legal or institutional requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Research ethics and consent facts and keep your reasoning.

A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Research ethics and consent facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Research ethics and consent to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled research ethics and **consent evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of research ethics and **consent**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Research ethics and consent

> Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation. Explain the labelled step before checking the changed condition.

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

Consent covers this study only. It excludes public posting of originals and unrelated face-recognition research.

AI ACADEMY / AI-7/19

Research ethics and consent

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State the purpose

Explain the drawing task and planned use before participation.



2. Respect exclusions

Do not publish originals or transfer them to the unrelated project.



3. Keep the agreement

Record permitted uses and the approved withdrawal procedure.

CHANGE ONE THING

A new project requests the same drawings for an excluded purpose.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Identify which uses the drawing **consent** permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record permitted uses and the approved withdrawal procedure.

1. State the purpose

2. Keep the agreement

3. Respect exclusions

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep the agreement Record permitted uses and the approved withdrawal procedure.

Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A new project requests the same drawings for an excluded purpose. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Do not reuse them under the existing **consent**; seek the required new authorisation.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Do not reuse them under the existing **consent**; seek the required new authorisation. The new purpose falls outside the stated agreement.

Repair a missing building block

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Research ethics and consent picture from its labels, then record a relevant limit.

01. Research ethics and consent: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

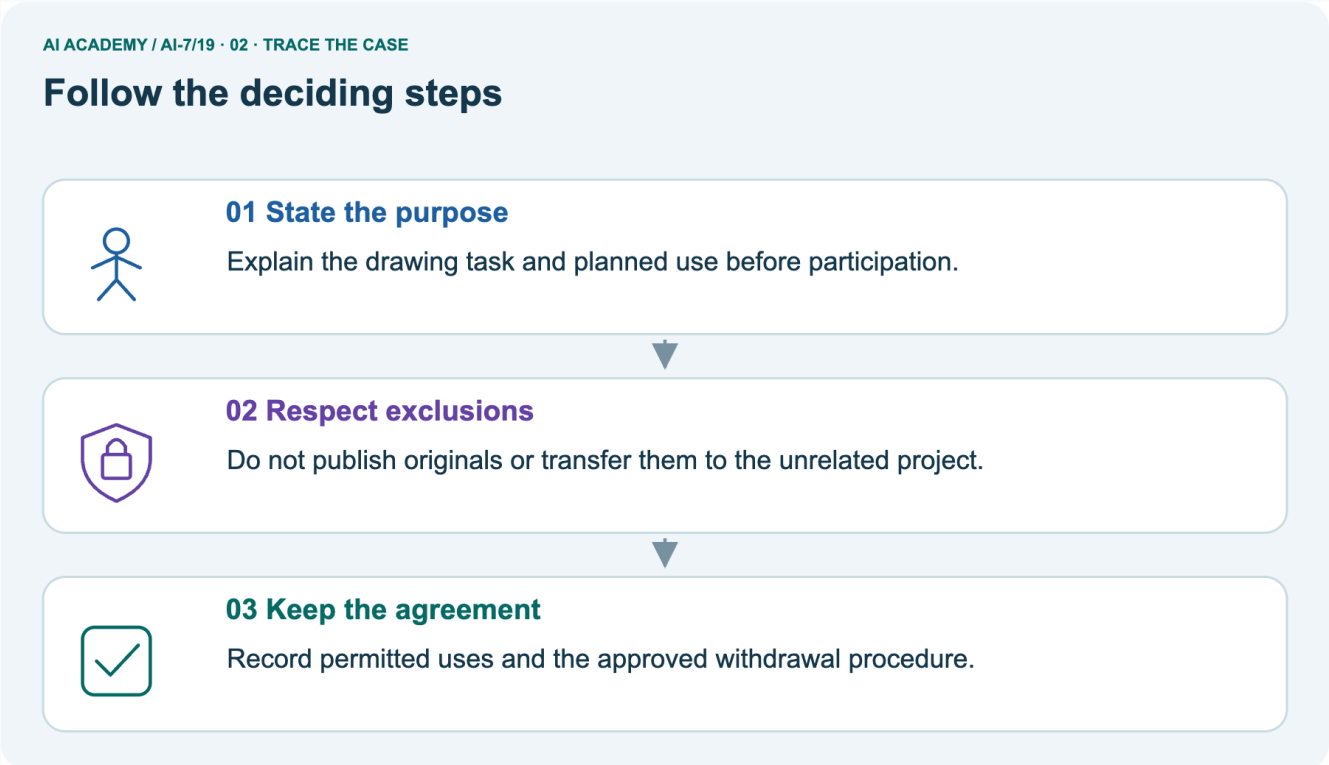
Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation. Supplied facts: **Consent** covers this study only.

It excludes public posting of originals and unrelated face-recognition research.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State the purpose: Explain the drawing task and planned use before participation. → Respect exclusions: Do not publish originals or transfer them to the unrelated project. → Keep the agreement: Record permitted uses and the approved withdrawal procedure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Research ethics and consent — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Research ethics and **consent** — **evidence** trace. Proposal automatically copies student essays into a study because they were submitted for grading. | Use fictional records for this course.

For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in **consent** and withdrawal handling before collection.

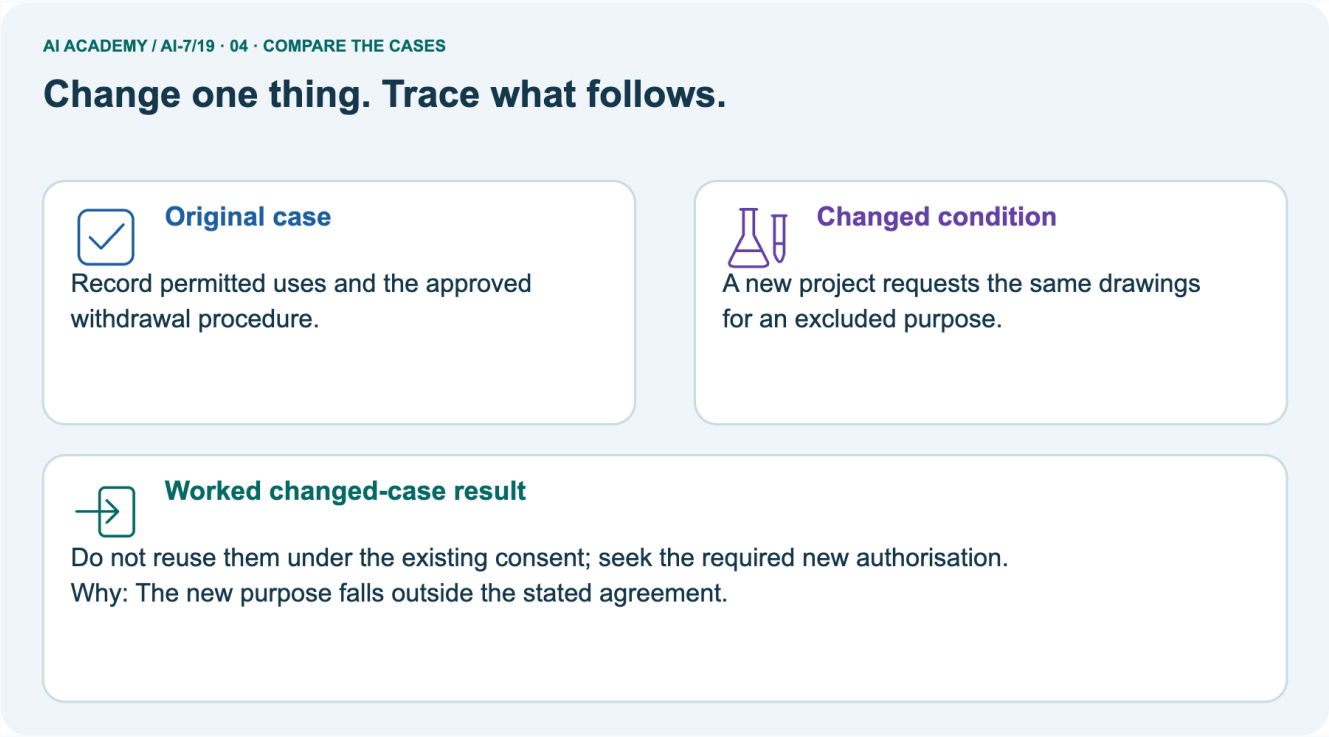
Do not implement real recruitment in this exercise. | The grading purpose does not establish research permission under the policy.

Redesign around fictional data or obtain required review and voluntary permission before collection.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

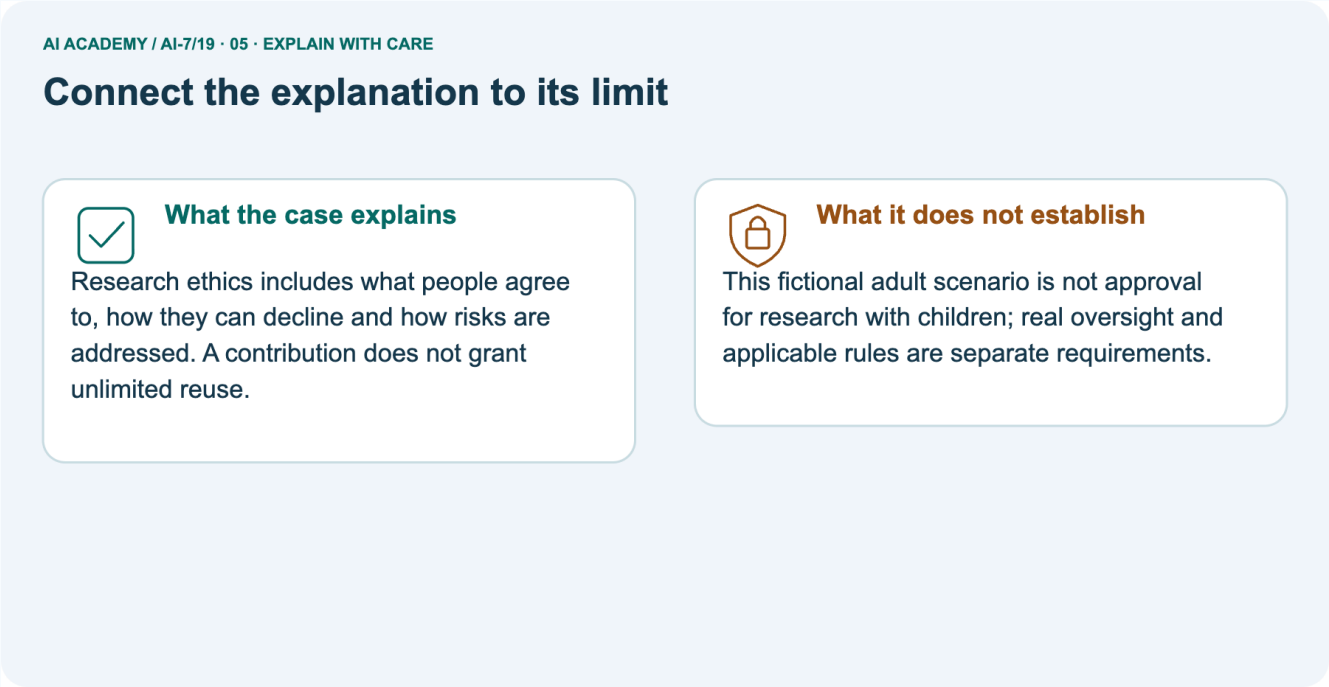
Original case: Record permitted uses and the approved withdrawal procedure. Changed condition: A new project requests the same drawings for an excluded purpose.

Expected result: Do not reuse them under the existing **consent**; seek the required new authorisation. Reason: The new purpose falls outside the stated agreement.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

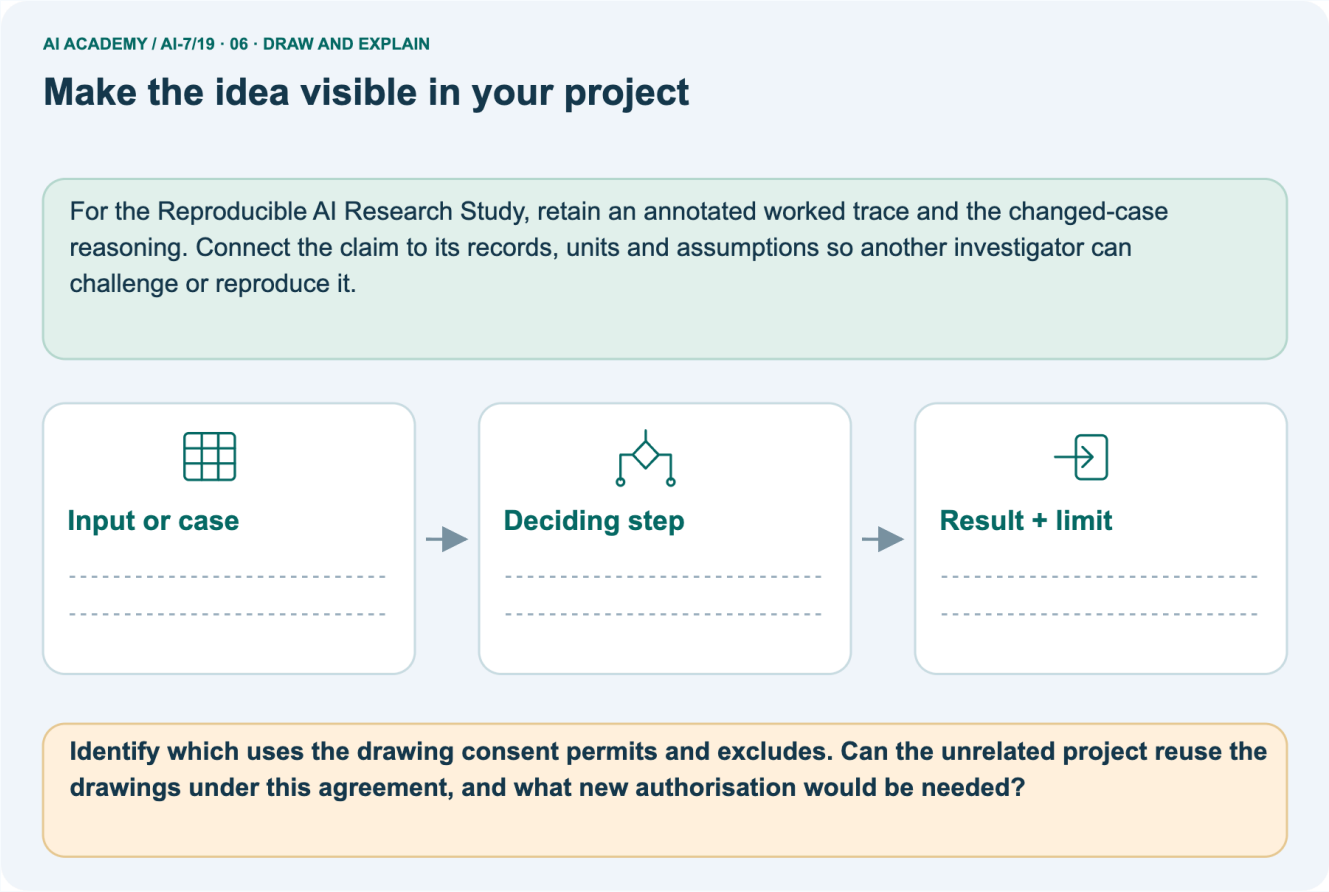
Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Limit: This fictional adult scenario is not approval for research with children; real oversight and applicable rules are separate requirements.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Identify which uses the drawing **consent** permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.